ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION WINTER SEMESTER, 2024-2025

DURATION: 2 HOURS FULL MARKS: 120

CSE 4107: Structured Programming I

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. a) Explain the rules for identifiers in C. Provide at least 4 different examples of invalid identifiers and explain why they are invalid. 4 + 4 (CO1) (PO1)

b) What output is produced by the code given in Code Snippet 1? Use '_' (underscore) to represent any whitespaces that appears in the output. 6 (CO1) (PO1)

```
1  #include <stdio.h>
2
3  int main(){
4      printf("%-6.3d, %5d\n", 35, 1024);
5      printf("%12.5e\n", 30.253);
6      printf("%8.4f\n", 83.162);
7      printf("%.4g\n", 123.45678);
8      printf("%5.3o\n", 32);
9
10     return 0;
11 }
```

Code Snippet 1: Sample program for Question 1.b

c) Write a C program that keeps taking integers as input until 0 is encountered (do not consider 0 as one of the inputs), and finally prints the largest and smallest integers among them. 10 (CO3) (PO1)

d) "One of the strengths of the C language is its flexibility, but simultaneously it is also one of its weaknesses" — explain this statement. 6 (CO2) (PO2)

2. a) Write a program that reads an integer number (as a single integer, not individual digits separately) from the user and finds the difference with its reverse number. A number's reverse can be obtained by writing its digits in the opposite order. For example: reverse of 452 would be 254. 12 (CO3) (PO1)

A few sample input-output pairs are given in Table 1.

Table 1: Sample input-output pairs for Question 2.a

Sample Input	Sample Output
452	198
121	0
21	9

b) Find the output of the code given in Code Snippet 2 and justify the result by showing in details whether each of the `if` conditions will be evaluated as `TRUE` or `FALSE`. 10 (CO2) (PO2)

```c
#include <stdio.h>

int main(){
    int a=1, b=20, c=5, d=3;

    if(a>=1 && b>c){
        printf("This\n");
    }
    if(b-- > 19 || c==d){
        printf("is\n");
    }
    if(d<a || ++c == 6 && d<c/2){
        printf("not\n");
    }
    if(!(c<d && a<=d) || b<c){
        printf("easy\n");
    }
}
```

Code Snippet 2: Sample program for Question 2.b

c) Write a C program that takes a positive integer as input and prints the factorial of that integer. Assuming that you used a `short int` data type to store the factorial result, determine the largest input for which the program correctly prints the factorial. 6 + 2 (CO3) (PO1)

3. a) Refer to the variable declarations as given in Code Snippet 3: 12 (CO1) (PO1)

```c
#include <stdio.h>

int main(){
    char c = '\2';
    short s = 2;
    int i = -3;
    long m = 5;
    float f = 6.5f;
    double d = 7.5;
}
```

Code Snippet 3: Sample program for Question 3.a

Determine the value and data type for each of the following expressions:

i. `c*i`
ii. `s+m`
iii. `f/c`
iv. `d/s`
v. `f-d`
vi. `(int)f`

b) Write a C program that asks the user to enter an integer n, and then prints the square of all odd numbers, say $oddSquares$, where $1 \leq oddSquares \leq n$. 10 (CO3) (PO1)

Table 2: Sample input and output squares for Question 3.b

Sample Input	Sample Output
Enter the upper limit: 150	1 9 25 49 81 121

c) What are meant by *lvalue* and *rvalue* in C? Explain with valid examples. 8 (CO1) (PO1)

4. a) Refer to the erroneous code given in Code Snippet 4 and the expected input-output given in Table 3: 10 (CO2) (PO2)

```
1  #include <stdio.h>
2
3  int main(){
4      int n;
5      scanf("%d",&n);
6      for(int i=0;i<n;i++){
7          for(int j=0;j<n;j++){
8              if(j%2!=0)
9                  continue;
10             printf("%d",j);
11         }
12     }
13 }
```

Code Snippet 4: An erroneous program for Question 4.a

Table 3: Sample input and Expected output for Question 4.a

Sample Input	Expected Output
3	1 333
6	1 333 55555
8	1 333 55555 7777777

The program was supposed to take a positive integer `n` as input and the expected output of was to print all odd numbers from `1` to `n` in separate lines. Also, each number would be printed as many times as its own value. Find the errors in the code given in Code Snippet 4 and fix the code with minimal changes.

b) "In C language, `v += e` is not equivalent to `v = v + e`" — justify this statement. 8 (CO1) (PO1)

c) Write a C program that asks the user to enter a date in the format `dd-mm-yyyy`, and then prints the date in the new format as shown in the sample input-output pairs in Table 4. Assume that the input is always a valid date. You must use switch-case to solve this problem. 12 (CO3) (PO1)

Table 4: Sample input-output pairs for Question 4.c

Sample Input	Sample Output
11-12-2013	the 11th of December, 2013
31-05-2024	the 31st of May, 2024
03-02-2001	the 3rd of February, 2001